PHY-112 | Principles of Physics-2

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Spring 2025 | Class #16

Department of Mathematics & Natural Sciences

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Enter Magnetic Field Intensity

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Where does it come from?

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Magnetic Field Sources

Only what we shall care about in this course

Magnetism is an interaction between moving charges. One creates it; another experiences it. But both must be ***moving***.

- A Magnetic field $\vec{B}$ is created by a moving charge
- Magnetic interactions are understood in terms of Magnetic poles: North and South
- Practical Magnetic fields are created by electric currents—collections of moving charges (Check Ørsted's Experiment).

We will not bother with permanent magnets in this course for Spring 2025.

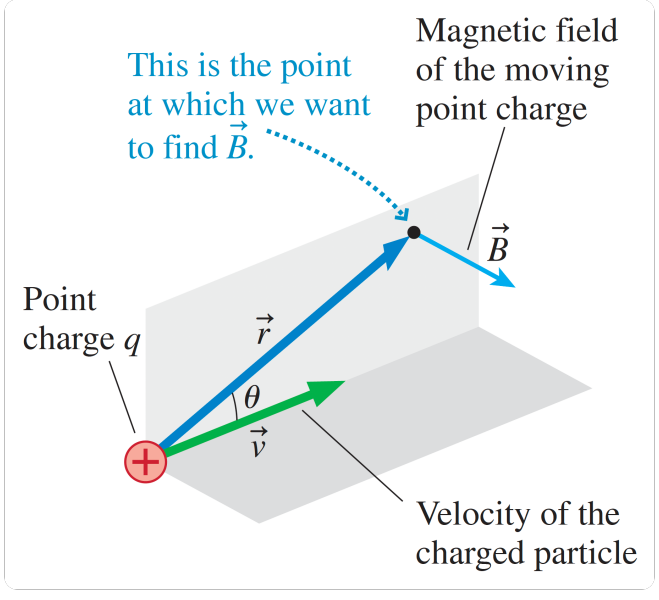
Calculate the Field intensity

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The Source of the $\vec{B}$-Field: Moving Charges

Creates circularly closed field around it

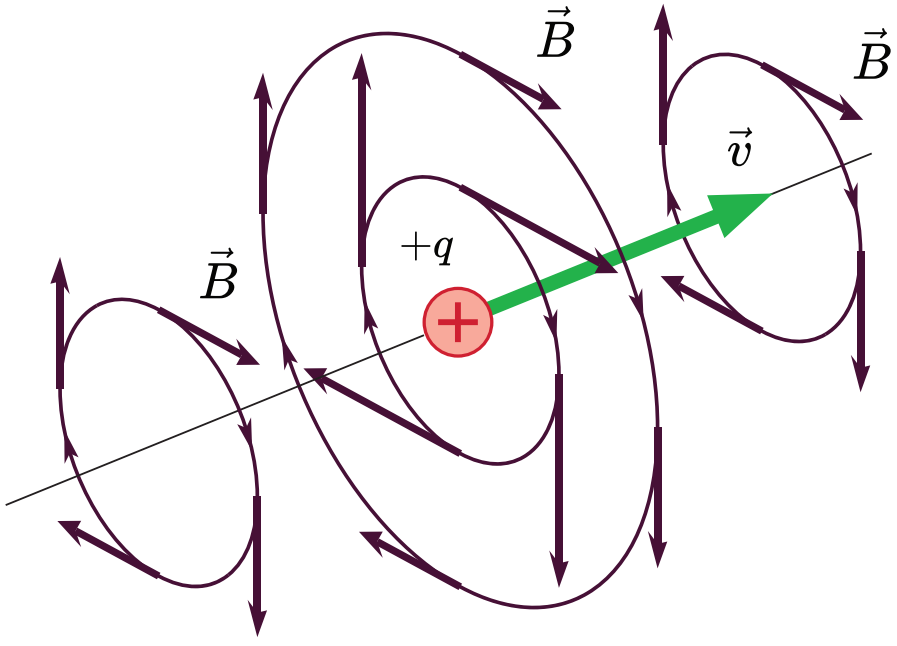
Biot-Savart (pronounce, *Bee-oh-Savah*) law measures the $\vec{B}$-field produced by a moving point charge. (Check how magnetic field lines form around a moving charge)



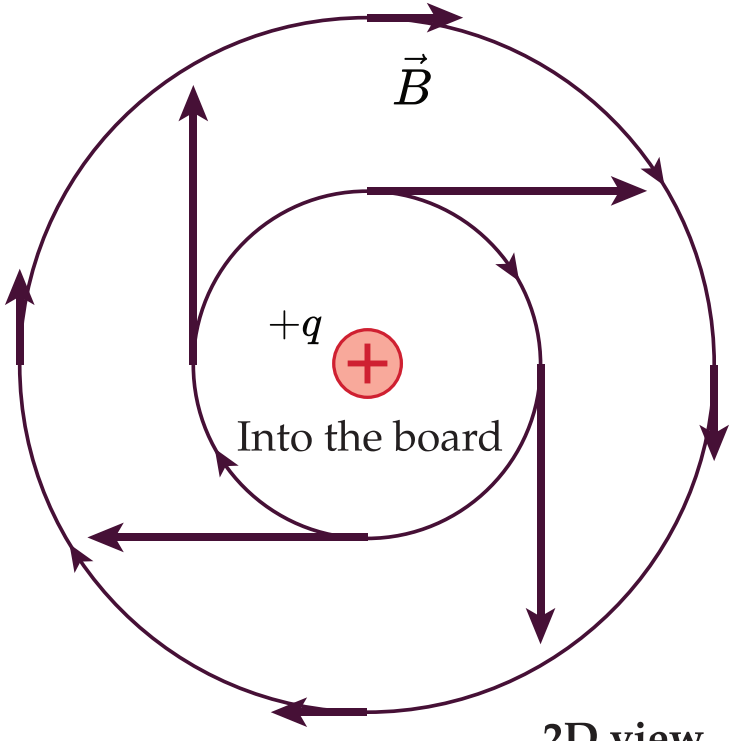
$$|\vec{B}|_{\text{point-charge}} = \frac{\mu_0}{4\pi}\left|\frac{q\vec{v}\times\hat{r}}{r^2}\right| = \frac{\mu_0}{4\pi}\frac{qv\sin\theta}{r^2}$$

$\vec{B}$-field lines for Single *moving* (positive) Point Charges

They are circular in 3D



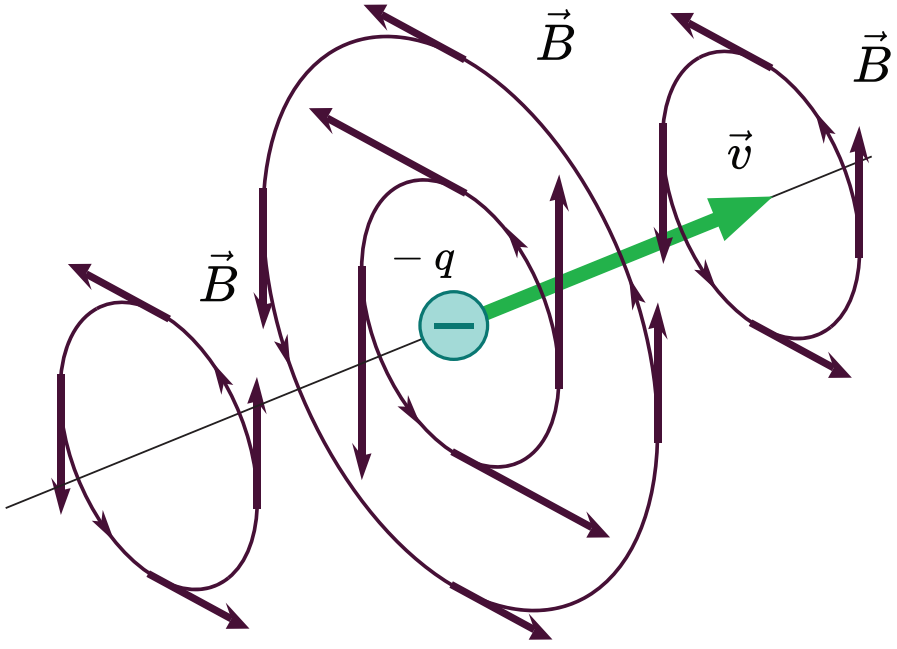
3D view



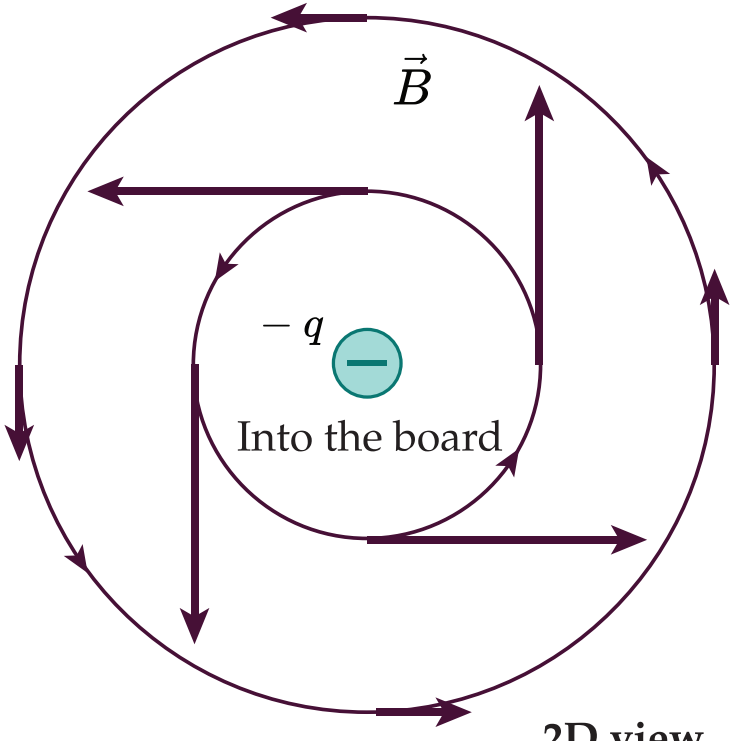
2D view

$\vec{B}$-field lines for Single *moving* (negative) Point Charges

They are circular in 3D



3D view

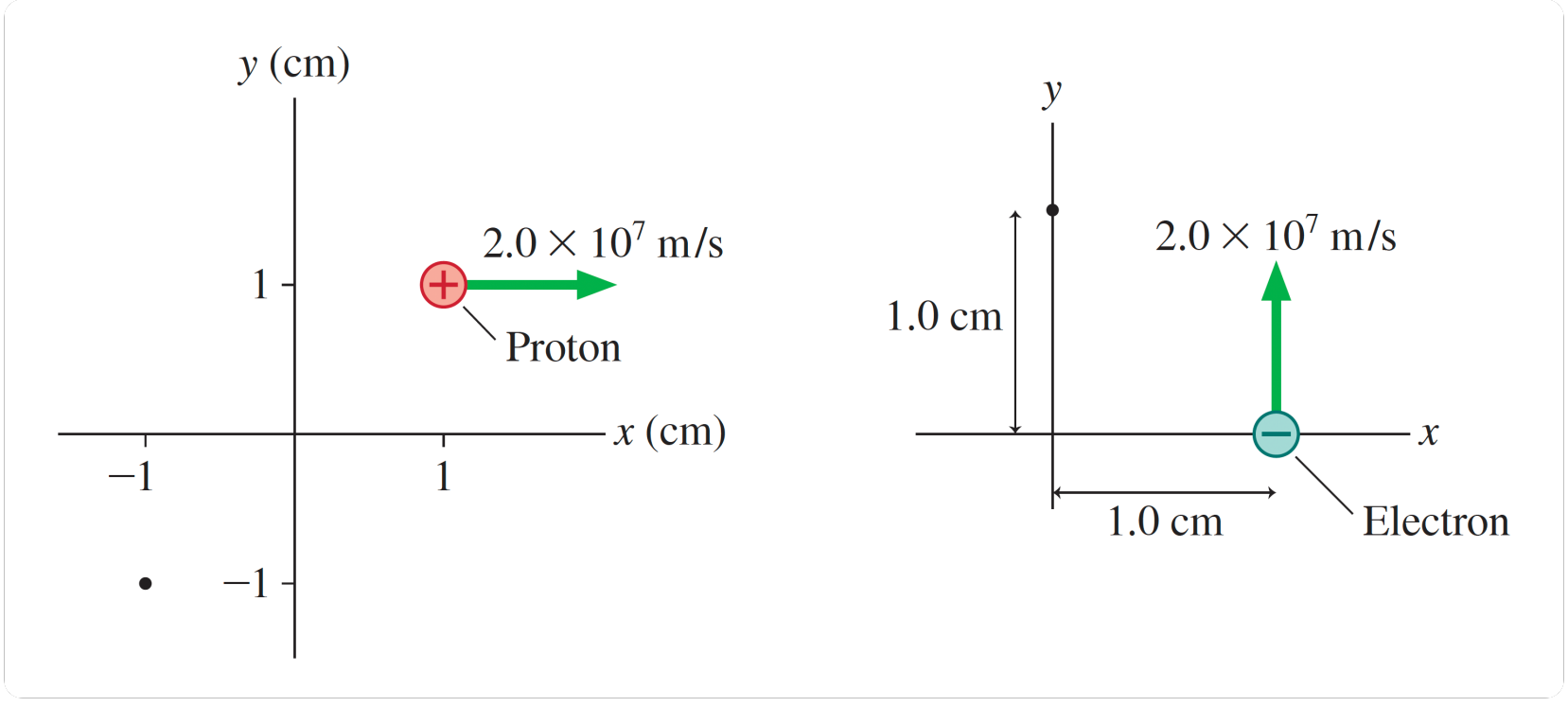


2D view

Incepting Ideas (1)

Hint: Utilize cross-product of two vectors calculation

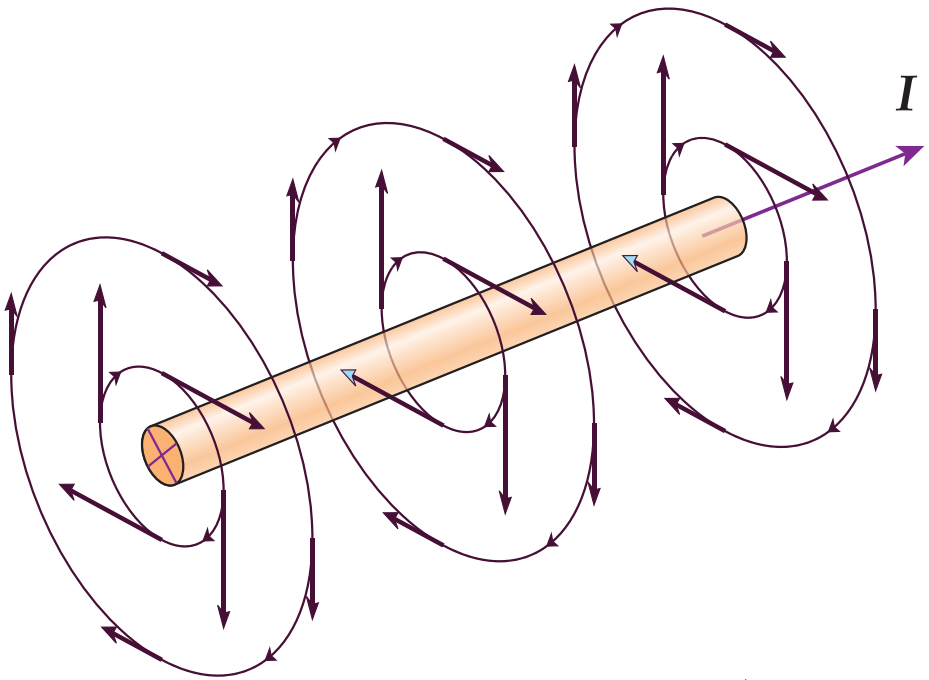
Q: What is the Magnetic field at the position of the dot in both diagrams? Give your answer as a vector.



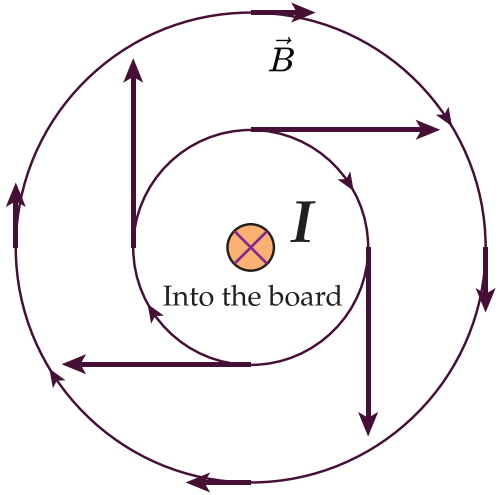
$\vec{B}$-field lines for Current-Carrying Wires

They are circular in 3D

3D view



2D view



$$|\vec{B}|_{\text{current-segment}} = \frac{\mu_0}{4\pi}\int\left|\frac{Id\vec{s}\times\hat{r}}{r^2}\right| = \frac{\mu_0}{4\pi}\int\frac{Ids\sin\theta}{r^2}$$

Incepting Ideas (2)

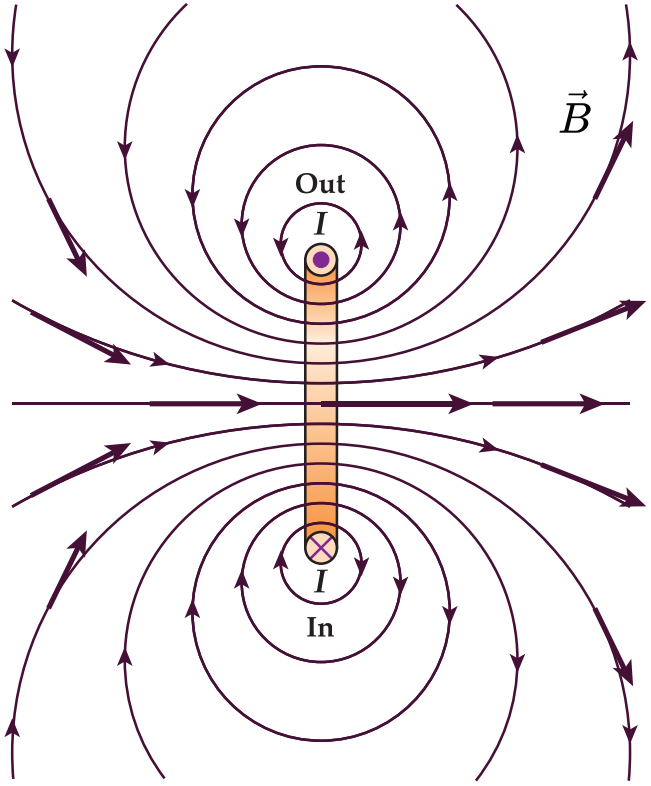
Hint: Utilize cross-product of two vectors calculation

Q: A small current element at the origin has a length of 2.0 mm and carries a current of 2.0 A in the $+z$ direction. Find the magnitude of the magnetic field due to this current element and indicate its direction on a diagram at (a) $x = 2.0$ m, $y = 4.0$ m, $z = 0$ m and (b) $x = 2.0$ m, $y = 0$ m, $z = 4.0$ m.

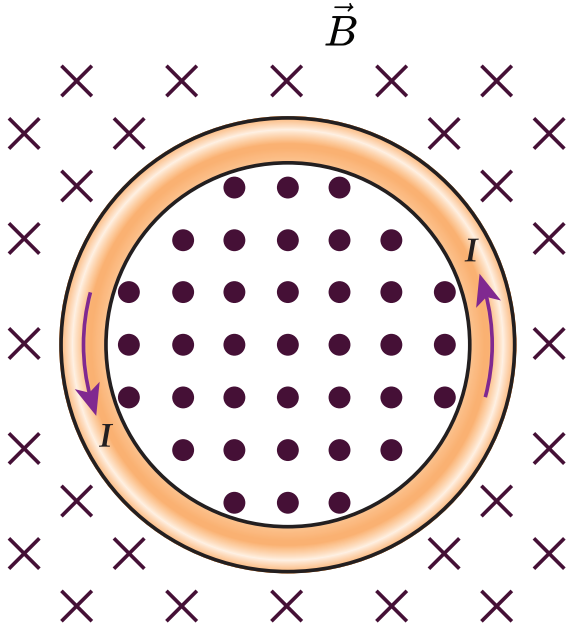
Three Important $\vec{B}$-field Models: Current Loop

Looks similar to the electric dipole

3D view (Cross-Section)



2D view



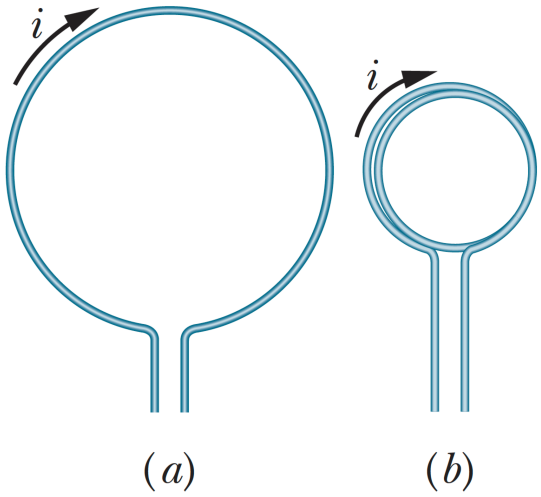
$$|\vec{B}|_{\text{current-loop}} = \frac{\mu_0 I}{2R}$$

$$|\vec{B}|_{\text{current-loop-with-turns}} = \frac{N\mu_0 I}{2R}$$

Incepting Ideas (3)

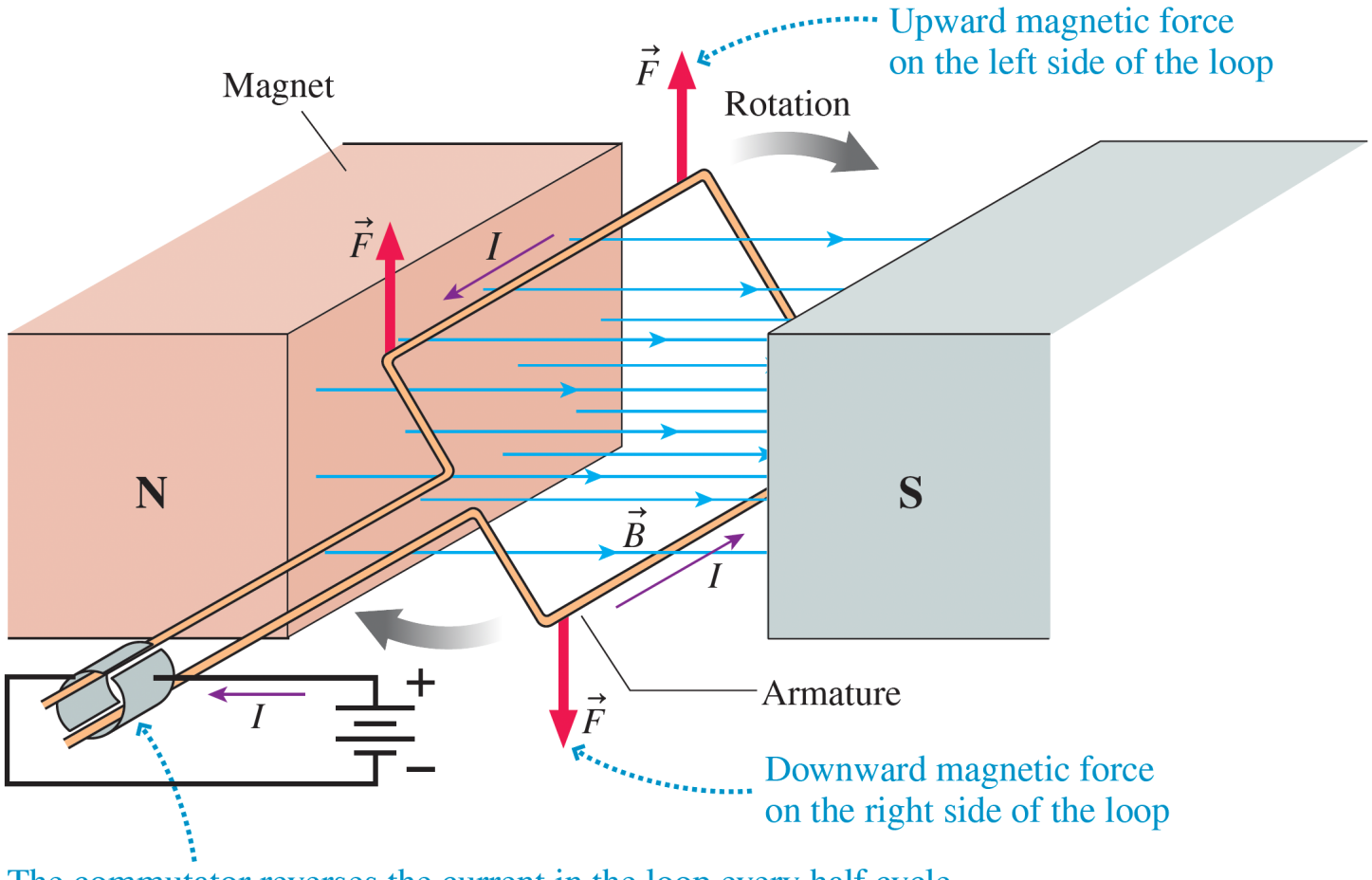
Hint: Use the expressions for B and μ for coils

Q: A length of wire carrying a current i and bent into a circular coil of one turn. The same length of wire has been bent to give a coil of two turns, each of half the original radius. (a) If B_a and B_b are the magnitudes of the magnetic fields at the centers of the two coils, what is the ratio $\frac{B_b}{B_b}$? (b) What is the ratio $\frac{\mu_b}{\mu_b}$ of the dipole moment magnitudes of the coils?



Yeah! But why should You care?

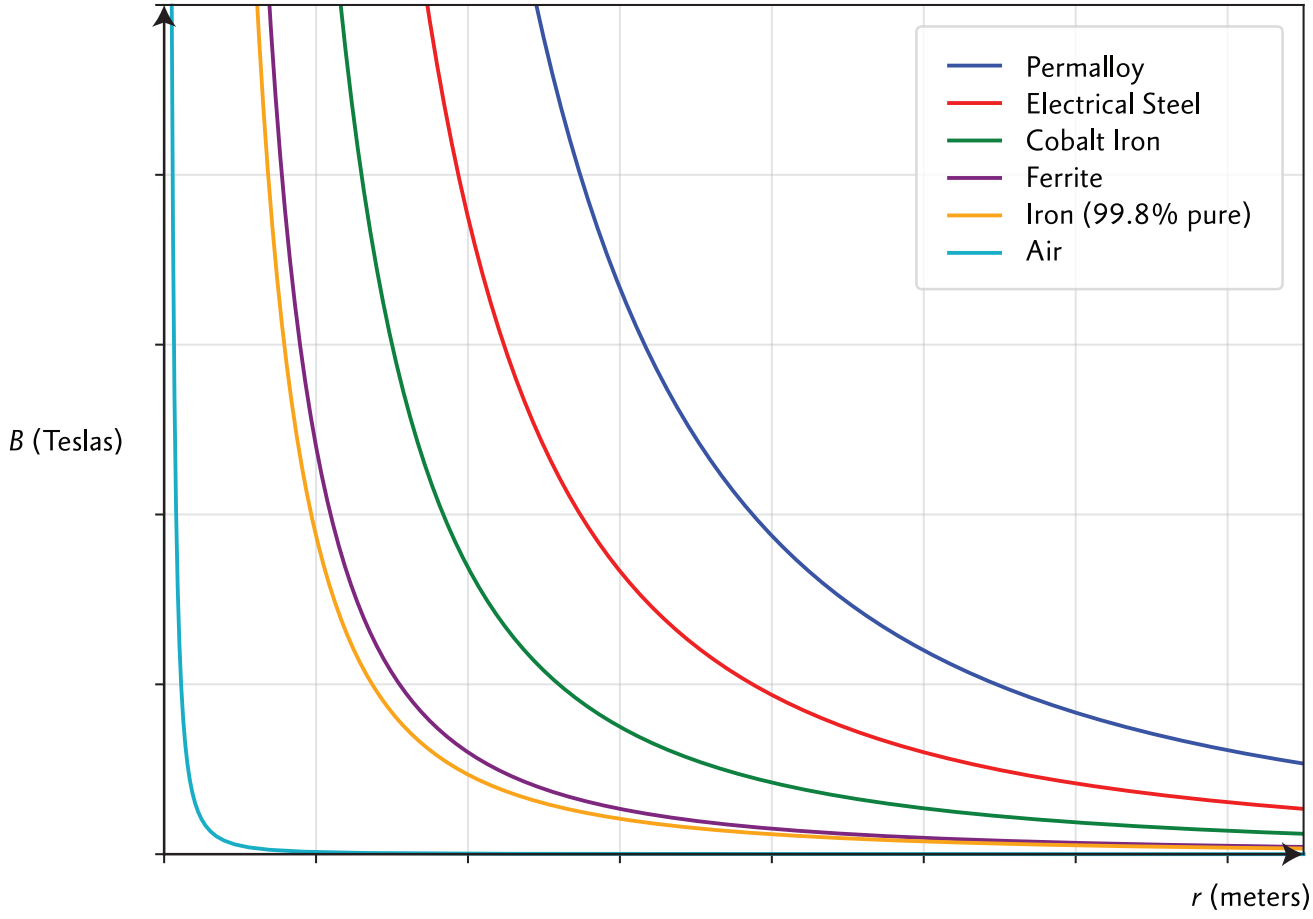
How to create an electric motor?



Effect of Permeability

We compare it with permeability of free space

Permeability of a medium is $\mu = \mu_r \mu_0$; where $\mu \rightarrow$ is the relative permeability compared with μ_0 permeability of free space.



Ampère's Law: Line Integration of $\vec{B}$-fields

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Ampère's Law: Physical Significance

Why do we use it?

What it is:

- It determines the Line integral of $\vec{B}$-field across a chosen path around a source
- It determines the Curl of $\vec{B}$-field in a given region
- It utilizes the high symmetry of the source to make the calculation simpler, similar to what Gauss's Law did for $\vec{E}$-fields
- It holds as long as the currents are steady and continuous.

Ampère's Law: The steps to follow

How do we use it?

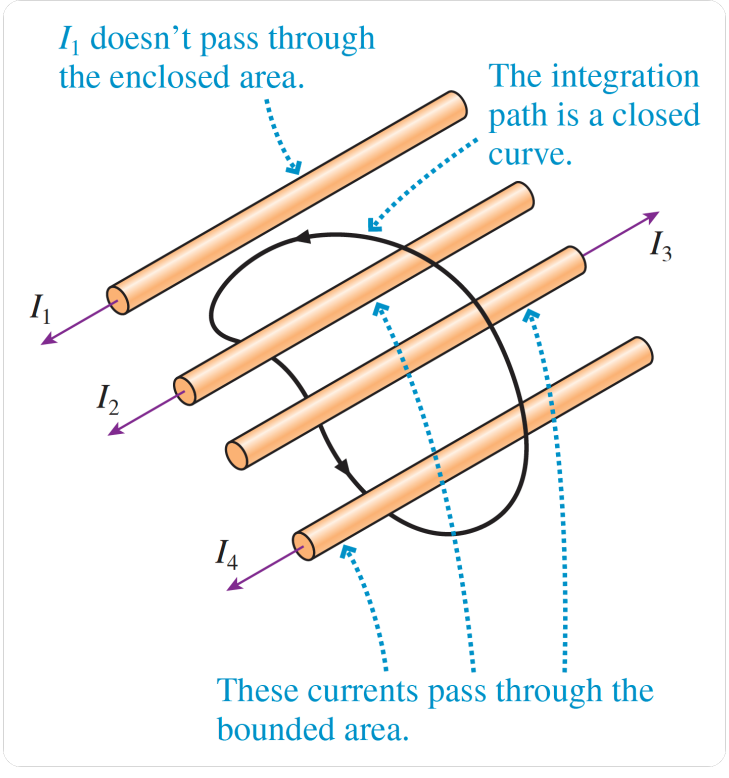
How it works:

- You choose a $\vec{B}$-field with high symmetry
- You choose a closed loop, called an *Ampèrian Loop*, on top of which the intensity does not change; only the direction does
- You take the line integral of $\vec{B}$-field in question across the closed loop
- You set it to $\mu_0 I_{\text{enc}}$, the algebraic sum of all currents within that loop
- Sign conventions for line integration help You calculate with no error

Sign Convention for Line Integration

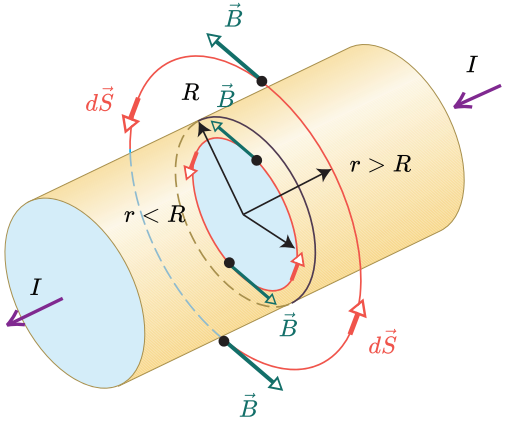
How to do the integration without losing your mind

Curl your four fingers in the direction of the integration, and you should have a unit vector $\hat{n}$, directed by Your thumb finger. Currents sharing $\hat{n}$'s direction shall be taken as **positive**. Any current outside of the loop does not factor into the calculation.

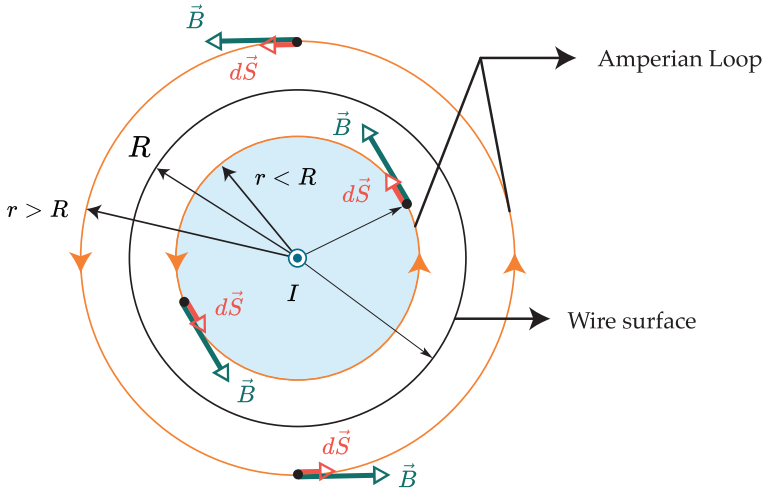


Ampère's Law in Action: Current Distribution

Easy way to find the field instensity



Side view

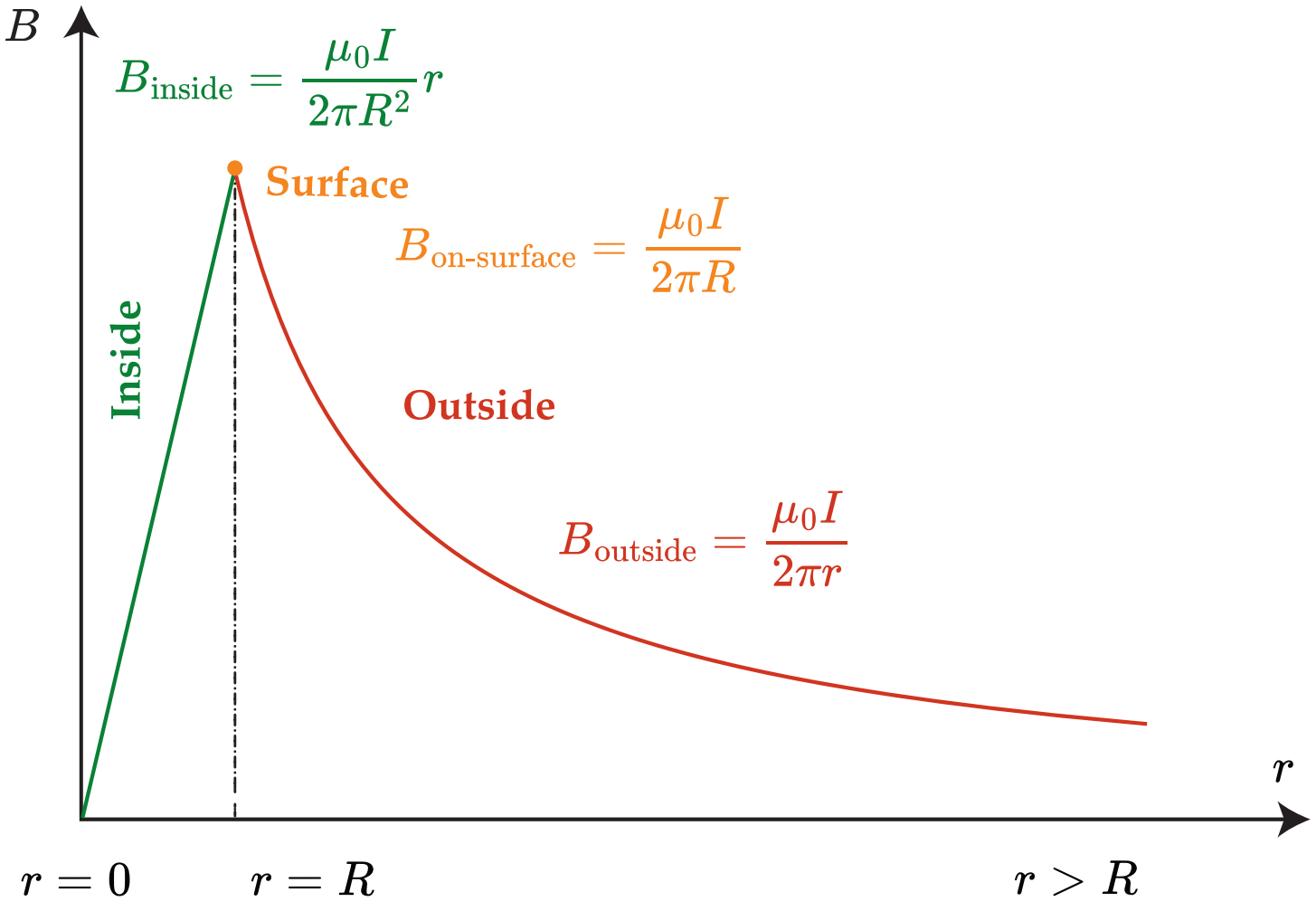


Top view

$$B = \begin{cases} \dfrac{\mu_0 I}{2\pi R^2} r & \text{for } r \leq R \\ \dfrac{\mu_0 I}{2\pi R} & \text{for } r = R \\ \dfrac{\mu_0 I}{2\pi r} & \text{for } r \geq R \end{cases}$$

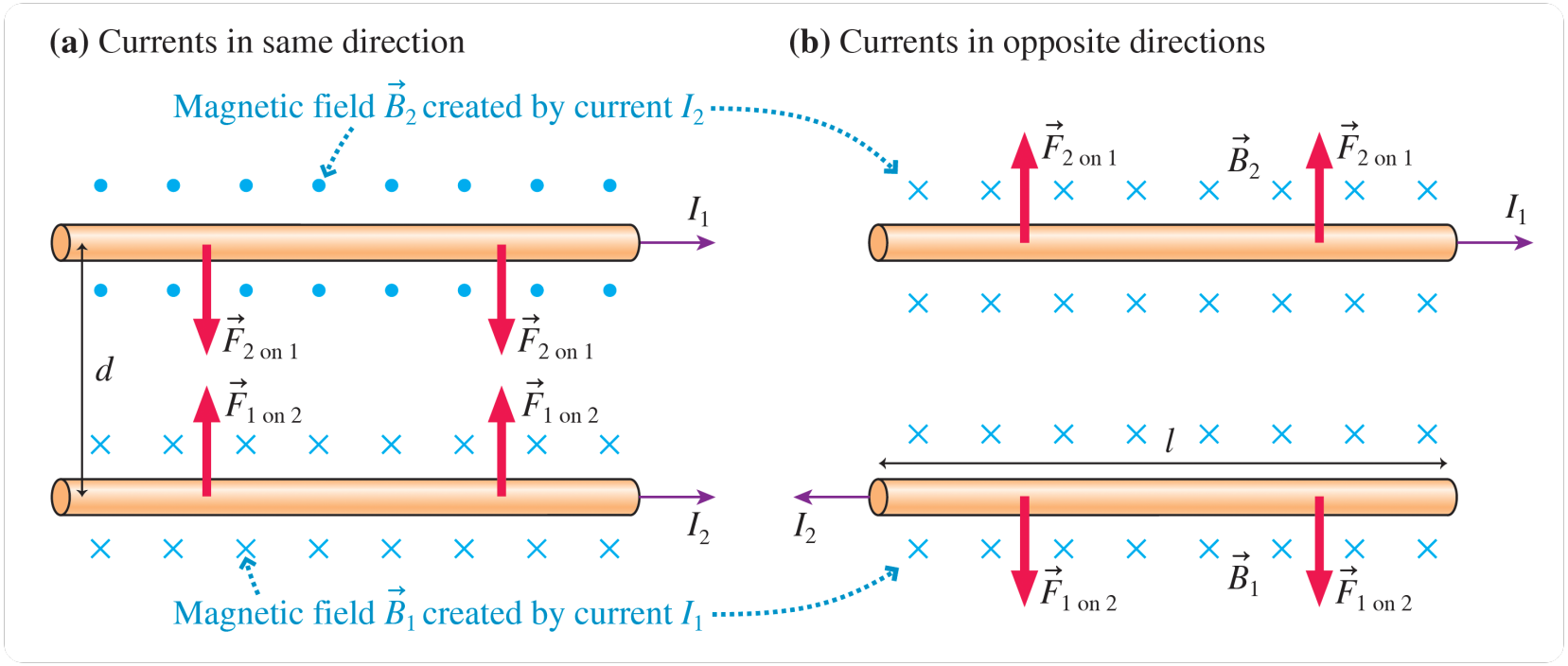
$\vec{E}$ Field for a Current Distribution

Ampère's Law at work where Bior-Savart failed!



Revisiting Parallel Current-Carrying Wire Setup

How to find the mutual forces. Finally!



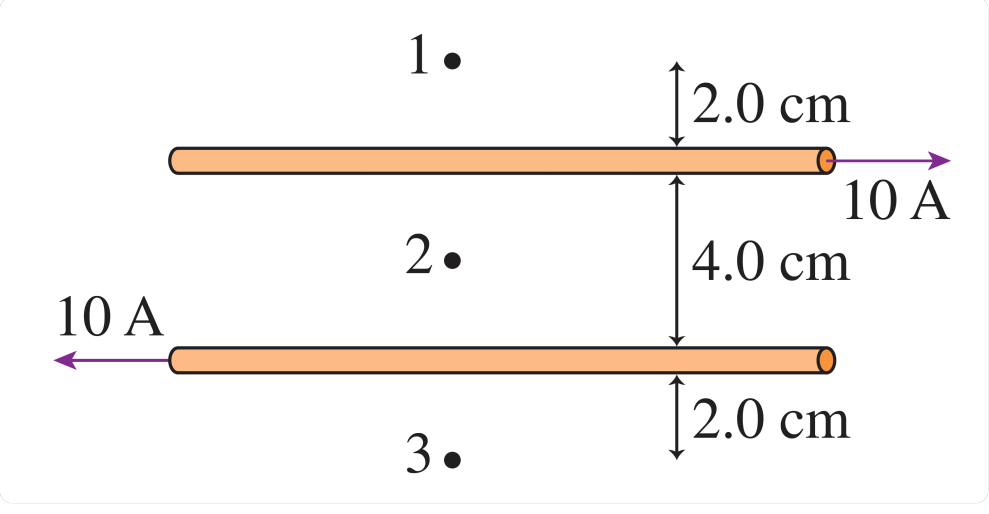
$$F_{1\,\text{on}\,2} = I_1 L B_2 = I_1 L \frac{\mu_0 I_2}{2\pi d} = \frac{\mu_0 I_1 I_2 L}{2\pi d}$$

$$F_{2\,\text{on}\,1} = I_2 L B_1 = I_2 L \frac{\mu_0 I_1}{2\pi d} = \frac{\mu_0 I_1 I_2 L}{2\pi d}$$

Incepting Ideas (4)

Hint: Superpose the field vectors

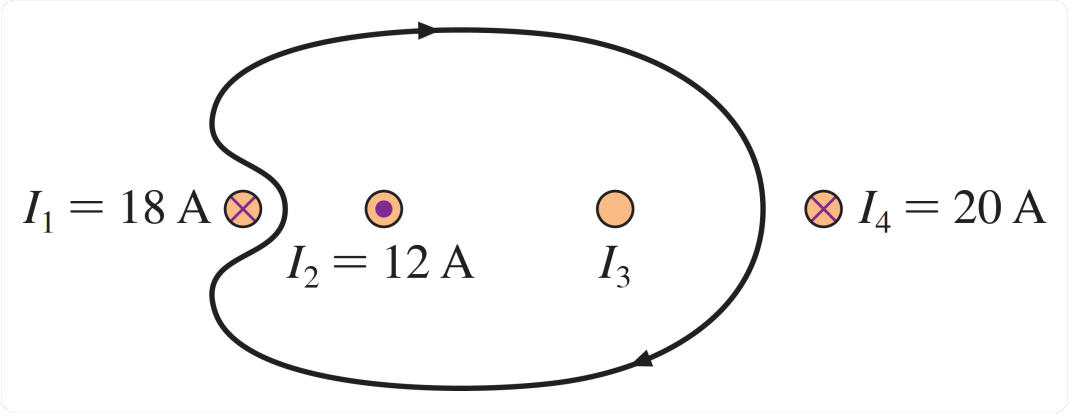
Q: What are the magnetic field strength and direction at points 1 to 3?



Incepting Ideas (5)

Hint: Follow the line integration convention

Q: The value of the line integral of $\vec{B} \cdot d\vec{s}$ around the closed path is 1.38×10^{-5} T m. What is the direction (into or out of the figure) and magnitude of I_3?



Remember Gauss's Law?

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Maxwell's Equation #2 in its full glory

Spoiler Alert: It is Gauss's law for Magnetic Fields

- The total Φ_B passing through a closed surface is always zero.

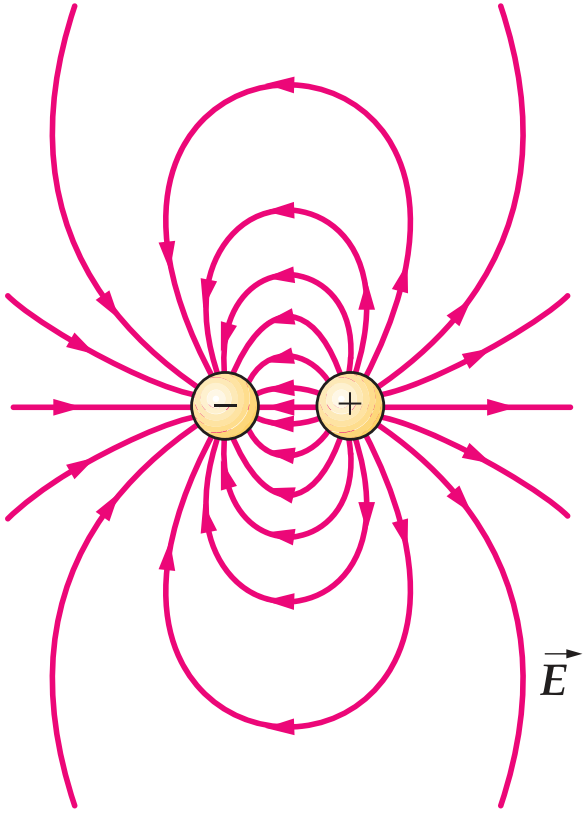
$$\oint \vec{B} \cdot d\vec{a} = 0. \qquad \text{(Integral Form)}$$

- The Divergence of $\vec{B}$-fields through a closed surface is always zero.

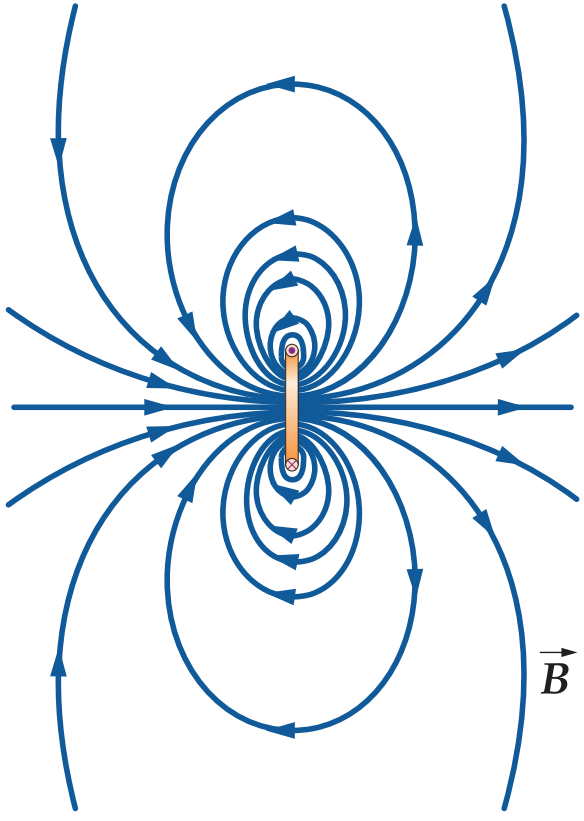
$$\vec{\nabla} \cdot \vec{B} = 0. \qquad \text{(Differential Form)}$$

No such thing as Magnetic Charges

Gauss's Law disapproves the existence of magnetic charges



Electric field lines
around an electric dipole



Magnetic field lines
around a magnetic dipole

Magnetic Flux

Measure the surface integral of $\vec{B}$

Q: What does electric flux measure?

- **Visually**: the number of magnetic field lines passing through a given surface.
- **Numerically**: the surface integral of $\vec{B}$-fields
  - $\Phi_B = \vec{B} \cdot \vec{A}$ (Uniform)
  - $\Phi_B = \int \vec{B} \cdot d\vec{A}$ (Non-Uniform)
  - Unit: [T m^2] or [Wb]
  - It is a scalar

Incepting Ideas (6)

Hint: Recall the flux expression for E, replace E with B

Q: Find the magnetic flux through a solenoid that is 40 cm long, has a radius of 2.5 cm, has 600 turns, and carries a current of 7.5 A. The magnetic field intensity of a solenoid is given by $B = \mu_0 nI$; where $n = \frac{N}{L}$ is the number of turns per unit length.

Overview on the Magnetic Field

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Overview on the Magnetic Field

Similarities/Differences between Electric and Magnetic Fields

Aspect	Electric Field	Magnetic Field
Monopoles/Isolated charges	Electric charges can exist in isolation with a diverging field	No such thing as magnetic charges It always comes in dipoles
Field-Based Interactions	Arises from electric field $\vec{E}$, governed by Maxwell's equations	Arises from magnetic field $\vec{B}$, governed by Maxwell's equations
Superposition	Fields from multiple sources add vectorially	Fields from multiple sources add vectorially
Maxwell's Equations	Governed by two of Maxwell's equations	Governed by two of Maxwell's equations
On-command field	Current-carrying wire aka a line charge, can produce $\vec{E}$-field, manually controllable	Current-carrying loop/coil produces a $\vec{B}$-field that can be controlled
Source	Electric charges (static or moving)	Moving charges (currents) or changing electric fields
Field Lines	Diverge from positive to converge on negative charges	Form closed loops with no start or end
Conservative Field?	Yes	No
Force	$\vec{F}_E = q\vec{E}$	$\vec{F}_B = q\vec{v} \times \vec{B}$
Interaction	On both stationary and moving charges	Only on moving charges

That is it for today!

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